

USER MANUAL

Chhatri

छतरी · umbrella

Income cover for small Paytm merchants,
where the claim starts itself.

Build for India AI Hackathon, Track 2 (AI-powered financial journeys)

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Version of 3 October 2026 · describes the prototype in this repository

Everything shown is a prototype that runs on synthetic demo data.

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HOW TO READ THIS MANUAL

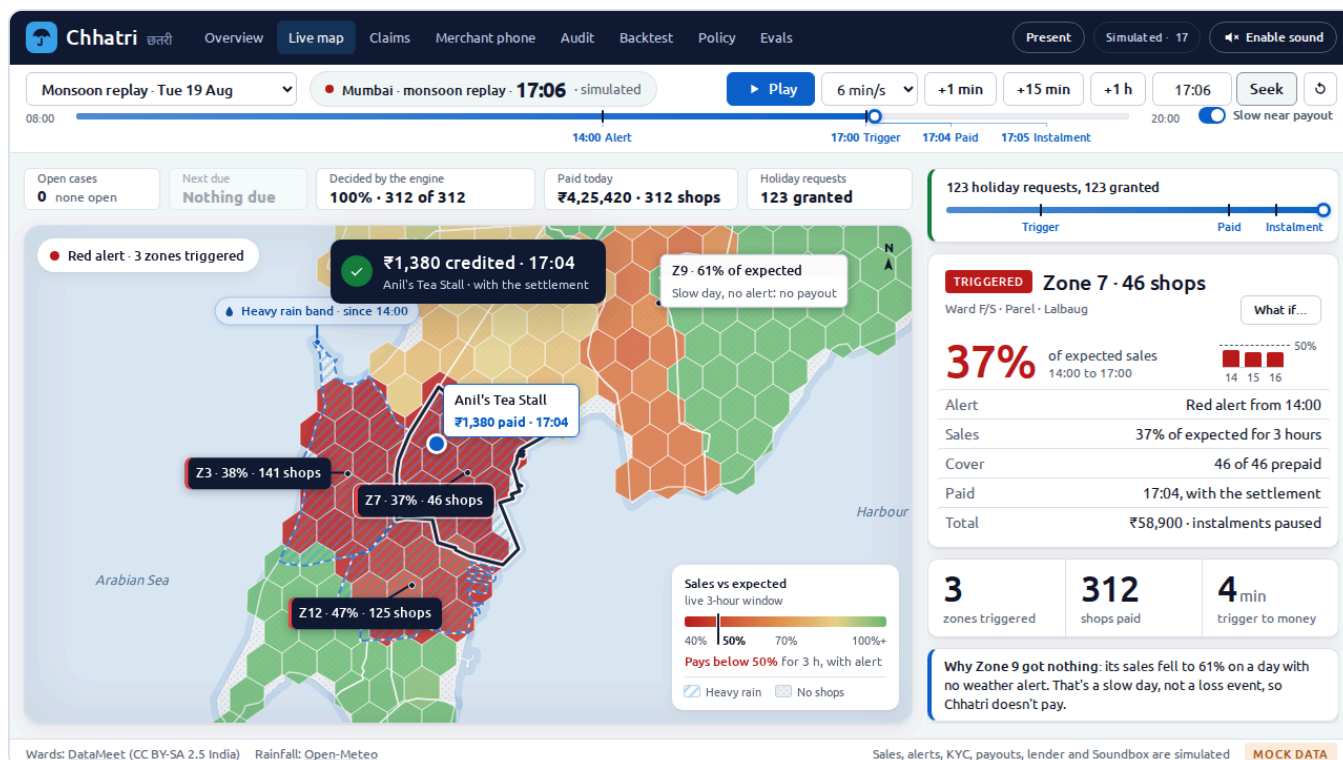
It assumes you know nothing about Chhatri. Every new word is explained the first time it appears, and there is a glossary at the end. If you only have five minutes, read section 1, then the three-minute story in section 7.

Words you will see on every page

Word	Plain meaning
Merchant	A small shop owner who takes payments on Paytm, such as a tea stall or a kirana (neighbourhood grocery).
Claim	A request to be paid because something bad happened. In Chhatri it often starts by itself.
Policy engine	The program that checks the rules and decides whether money is paid. It is ordinary code, not AI.
Simulated	Pretend but clearly labelled. For example, the WhatsApp phone is a drawing inside the console.
Demo flag	A switch that turns one feature on. All flags are off until you set them (section 5).

1 What Chhatri is

Chhatri (छहत्री, Hindi for "umbrella") is income cover for India's small merchants. When something stops a shop from earning, such as a red rain alert that empties the street, Chhatri notices the loss in the shop's own sales, pays the same day, explains the amount in Hindi, and asks the shop's lender to give that day's loan instalment a holiday. The merchant does not fill in a form. **The claim starts itself.** The AI builds the case and a rules engine written in ordinary code decides the money. Every step goes into a log that shows if anyone changed it.



The picture in one screen. It is the live map of Mumbai at 17:06 on the demo storm day. Red wards were paid automatically: 3 zones, 312 shops, 4 minutes from trigger to money. Anil's tea stall gets ₹1,380. Sales, alerts and payouts here are simulated and labelled as such.

THE ONE-LINE VERSION

Rain stops sales. Paytm sees the drop. Chhatri pays that evening, with no form and no phone call, and the lender decides about the instalment.

2 The problem, and who it is for

The problem

A tea stall earns day by day. On a day of heavy rain nobody stops for tea, and the stall's sales can fall by more than half. The owner still has to pay his loan instalment. Illness does the same: if the owner is in hospital, the shop is shut and no money comes in.

Insurance exists for such days, but it is slow. The facts file for this project records that earlier merchant protection plans on Paytm took **30 to 60 days per claim** and needed several documents (fact A3). A merchant living on daily sales cannot wait that long.

The story used in the demo shows the gap. Anil runs a tea stall in Parel, Mumbai. His usual Tuesday sales are **₹4,380**. On the storm day his area's sales fall **63%**. His loan instalment of **₹600** is still due. With Chhatri he is paid **₹1,380** the same evening and his lender is asked to pause the instalment.

Number	What it tells you	Where it comes from
1.57 crore	Merchants who paid for Paytm payment devices in Apr-Jun 2026. These are the people Paytm already sees the sales of.	Facts file, A1
30-60 days	How long earlier merchant protection plans took per claim, with several documents.	Facts file, A3
About 250-276 mm in about 6 hours	Heavy rain in Mumbai on 25-26 Sep 2024, the kind of day that empties a street.	Facts file, A13
About 1 in 3	Chance of no payout even when the area lost everything, in a study of 270 weather-index crop insurance products. This gap between the trigger and the real loss is called basis risk.	Facts file, A8
63%, ₹4,380, ₹1,380, ₹600	Anil's demo day: sales drop, usual Tuesday, payout, instalment.	docs/DEMO.md
3 zones, 312 shops, 4 min	Demo storm result: zones triggered, shops paid, minutes from trigger to money.	docs/DEMO.md

Who uses Chhatri

The merchant

A shop owner such as Anil. Uses a phone app in Hindi or English: sees cover, gets paid, asks questions, files a complaint. Does not file claims for a storm.

The claims officer

A person who looks at the doubtful cases: a slip with the wrong name, or a merchant who says the amount is too low. Works in the console on the Claims page.

The lender

The business that gave the merchant the loan. Chhatri only *asks* it to pause an instalment. The lender decides. In this prototype the lender is simulated.

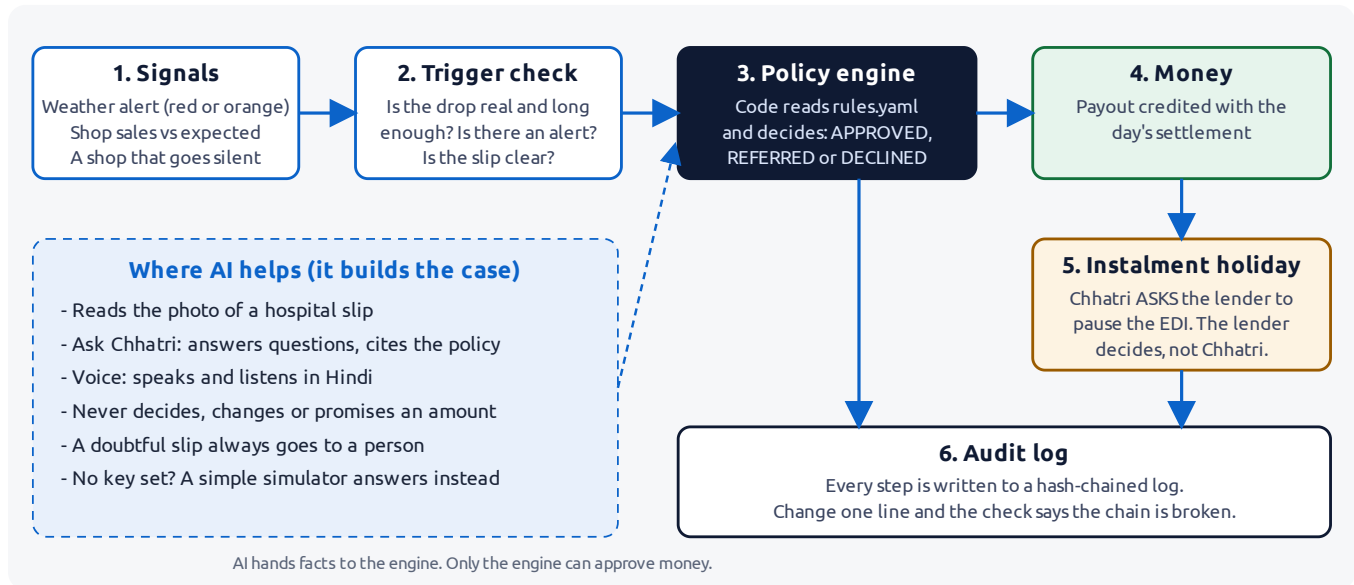
Why it matters

- **Speed.** Money arrives the same evening, not weeks later.

- **No paperwork for a storm.** An area claim needs no document at all. A hospital claim needs one photo of the hospital slip.
- **Fairness you can check.** Every amount is a sum the merchant can read: half of the usual day, times the drop.
- **Humans stay in charge of doubt.** If the AI is not sure, a person decides.

3 How it works

Follow one claim from left to right. A **trigger** (something that starts a claim) goes to the **policy engine**, which reads a rules file called `rules.yaml`. If the rules say yes, money is paid and the lender is asked about the instalment. Everything is written to the **audit log**.



The two kinds of claim

	Area claim (storm)	Personal claim (illness)
What starts it	A red or orange weather alert, and the hourly sales index of a zone stays below 50% of expected for 3 hours in a row, with at least 20 shops in the zone.	A shop shows a full day with no sales. Chhatri checks in on WhatsApp in Hindi. The merchant sends one photo of the hospital slip.
Documents	None.	One hospital slip.
Amount	Half of the lost sales, capped at ₹2,500 per shop per day.	Half of the usual day, capped at ₹1,500 a day, up to 3 days paid automatically.
Example	$\frac{1}{2} \times ₹4,380 \times 63\% = ₹1,380$ (Anil).	$\frac{1}{2} \times ₹4,300 = ₹2,150$, capped at ₹1,500 for 1 day = ₹1,500.

THE KEY WORDS

Expected sales: what the shop normally sells at this hour, estimated by a small statistical model. **Sales index:** actual sales divided by expected sales. **Zone:** a group of neighbouring shops in one area; the demo uses Z3, Z7, Z9 and Z12. **Settlement:** the daily payment Paytm makes to a merchant. **EDI:** the equated daily instalment of a loan, taken out of the settlement. **EDI holiday:** a pause of one instalment, which moves to the end of the loan.

Who is allowed to decide what

Case	Chhatri alone	Goes to a human
Area drop during an alert, index clear	Pays	Only if the merchant disputes
Personal claim, slip matches name and dates	Pays up to the daily cap	Anything above the cap

Case	Chhatri alone	Goes to a human
Slip unclear or dates do not match	Never	Always
Cover bought after an alert	Never	A waiting period applies

This is the table on the Policy page (screen 6.13). Other rules in `rules.yaml`, version `pilot-0.1`: payout share 50%; slip name must match the KYC name (score 85 or more); slip reading confidence 0.80 or more; waiting period 7 days; annual limit ₹30,000; disputes answered within 24 hours. **KYC** means the identity record of the merchant.

Where AI helps, and where it never decides money

AI helps with	How	It never
Reading a hospital slip	A vision model reads the name, dates and hospital. The merchant confirms what was read.	Approves a claim. The engine checks the name against KYC.
Ask Chhatri	Answers questions in plain words and cites the policy clause. Every number comes from the engine.	Invents an amount or promises a payout.
Voice	Listens and speaks in Hindi and English. Amounts and dates are confirmed with a tap.	Acts on an amount the merchant has not confirmed.
Hidden instructions	Text found inside slips and chat is treated as data, never as an order.	Obeys it.

THE RULE THAT MATTERS

The AI builds the case. Code decides the money. A message that tells a merchant an amount is filled from a fixed template with numbers from the engine, never written freely by a model.

4 What is real and what is simulated

Chhatri is a prototype, and it says so on screen. Every outside service has a small label in the console header (click the chip that reads *Simulated*). There are three labels:

Label	Meaning
LIVE	A real service is being called, because you set its key. Only AI services can be live here.
SIMULATED	A simulator answers instead. Nothing outside is called.
FALLBACK	The service was meant to be live but failed or was switched off, so a simpler answer is used. This label appears with the provider panel flag.

Part	Status in this prototype
Policy engine, forecast model, audit log, claims console	Real code, running for real (on simulated shop data).
Shop sales, weather alerts feed, KYC, payout rail, lender, Soundbox (the speaker on the till)	Always simulated and always labelled.
WhatsApp	Simulated. The phone you see is drawn inside the console. It becomes live only with four WhatsApp settings, which this project has not run.
Paytm payment link	Simulated. A pretend link such as <code>https://paytm.me/sim-...</code> is shown and labelled.
Sarvam (speech, chat, slip reading)	Simulated unless <code>SARVAM_API_KEY</code> is set.
Gemini (chat and slip reading)	Simulated unless <code>GOOGLE_API_KEY</code> and <code>GEMINI_MODEL</code> are set.
Workflows (n8n)	An in-process runner does the same steps. The n8n label reads simulated unless you run the Docker stack.
Weather	Rain data in the backtest is real, from Open-Meteo, saved in the repository. The live weather call is off by default.

HONEST LIMITS

- AI services run **only on synthetic demo data**. The setting `CHHATRI_DATA_IS_SYNTHETIC=true` is the declaration. Without it, every free-tier AI call is skipped.
- The backtest (section 6.12) uses **simulated sales with real rainfall**. It tests the rule, not accuracy on real shops. Its simulation was tuned so the demo numbers appear, and premiums are priced so each zone ends near a 65% loss ratio, so those numbers are by design, not a finding.
- No insurer, lender or Paytm team has agreed to take part. Nothing here is an approved insurance product.
- No measured accuracy for any live AI model exists yet. The evaluation page says NOT MEASURED until a run is stored.

5 Install and run

What you need first

- **Python 3.12** and **Node 22** (Vite, the web tool, needs Node 20.19 or newer) and **GNU make**.
- **Docker with Compose v2**, only if you use `make up`.
- No API key is needed. Without keys every service is simulated.

Run everything on your machine

```
git clone <repo> chhatri && cd chhatri
make setup      # backend/.venv with the backend, and npm ci in frontend/
make env        # creates .env with a generated secret (never overwrites an existing one)
make dev        # backend on http://localhost:8000, console on http://localhost:5173 (Ctrl+C stops both)
```

Open `http://localhost:5173`. The backend loads the **monsoon** scenario at start, paused at 08:00. `GET http://localhost:8000/api/preflight` lists what is ready. `make data` rebuilds the saved model files; it is slow and needed only if a file is missing, because the results are committed.

Run with no backend at all (the static mock)

The console can run a copy of the whole backend inside the browser. Nothing else needs to run, so it works offline and can be put on a free static host.

```
cd frontend
npm ci
npm run dev:mock                # development server in mock mode

# a static build, shown exactly with the flags it was built with:
VITE_FEATURES=n1_miniapp npm run build -- --mode mock
npm run preview                 # then open /merchant/S-0142/app
```

In mock mode every service is simulated and the footer shows a "Mock data" badge. A full page reload restarts the mock, so move around with the links, not the reload button.

Check that it works

Command	What it does
<code>make demo-check</code>	Runs every demo scenario through the HTTP API and prints <code>PASS: 70 passed, 0 failed</code> when all is well. It runs on its own copy of the app with only <code>x4_lender_request</code> on. Do not turn on <code>n3_slip_precheck</code> or <code>n6_consent</code> for it: the scripted steps stop by design when those two are on.
<code>make check-keys</code>	Prints whether <code>SARVAM_API_KEY</code> and <code>GOOGLE_API_KEY</code> are SET or NOT SET. It never prints a key. With a Google key it lists the Gemini models the key can use.
<code>make test</code>	Backend tests (coverage gate 80%) and frontend type check, lint and unit tests.
<code>make e2e</code>	Playwright browser tests against running servers.
<code>make up / make down</code>	Docker stack: backend, console on <code>http://localhost:8080</code> and <code>n8n</code> on <code>http://localhost:5678</code> .

API keys (optional)

Put keys in `.env` (the file `make env` creates). Leave them out and everything stays simulated.

Setting	Turns on
<code>SARVAM_API_KEY</code>	Sarvam speech to text, text to speech, chat and slip reading.
<code>GOOGLE_API_KEY</code> and <code>GEMINI_MODEL</code>	Gemini chat and slip reading. A key alone is not enough: a model id is needed too (<code>GEMINI_VISION_MODEL</code> is optional).
<code>CHHATRI_DATA_IS_SYNTHETIC=true</code>	The declaration that all data is synthetic. It is already in <code>.env.example</code> . Never set it where real merchants' or patients' data is held.
Four WhatsApp settings; <code>PAYTM_MCP_URL</code> or <code>PAYTM_MID</code> + <code>PAYTM_KEY_SECRET</code> ; <code>N8N_BASE_URL</code>	Live WhatsApp, a live Paytm staging link, n8n workflows. Not used in the demo.

The 14 feature flags

A **feature flag** is a switch. Every feature added for the final is off until you name it. Off means absent: the screen is hidden and the backend answers an ordinary 404. Set the same list in two places:

`CHHATRI_FEATURES` for the backend (in `.env`) and `VITE_FEATURES` for the console, separated by commas.

```
CHHATRI_FEATURES=n1_miniapp,n2_ask_chhatri,n3_slip_precheck,n4_voice,n5_grievances,n6_consent  
s,n8_marathi,x4_lender_request,x6_provider_panel,x8_distress_guard,h8_ops_strip,h24_whatif,h25_evals,  
console_polish  
# the console needs the same names:  
VITE_FEATURES=<the same list> npm run dev:mock
```

Flag	What it turns on
<code>n1_miniapp</code>	The merchant mini-app: home, coverage, buy cover, claims tracker, claim detail, why this amount, receipt, help, language switch (Hindi, English), jargon lens, next-step bar.
<code>n2_ask_chhatri</code>	Ask Chhatri: questions answered with policy clause citations and a scam-message warning.
<code>n3_slip_precheck</code>	The slip pre-check: the app shows what it read and a checklist, and the merchant confirms before any check runs.
<code>n4_voice</code>	Voice: the microphone and the confirm chips for amounts and dates.
<code>n5_grievances</code>	The complaint ladder with response clocks.
<code>n6_consent</code>	The consent centre, the consent activity log, "Erase this slip", and the consent tick boxes when buying cover.
<code>n8_marathi</code>	Marathi as a third language (a draft, text only).
<code>x4_lender_request</code>	The lender decides the EDI holiday (without it the 17:05 line is an unconditional pause). The demo flag.
<code>x6_provider_panel</code>	The provider panel with Force fallback switches and the FALLBACK label.

Flag	What it turns on
<code>x8_distress_guard</code>	No loan offers to a merchant in distress, and a cap on how many messages one merchant gets.
<code>h8_ops_strip</code>	The strip of five real operations numbers at the top of the console.
<code>h24_whatif</code>	The "What if..." drawer on a zone card.
<code>h25_evals</code>	The Evals page (AI evaluation).
<code>console_polish</code>	Presenter mode, the trigger-to-payout card and projector-sized type.

You can see which flags are on with `GET /api/health`. Run `make dev` or `npm run dev:mock` with all 14 flags on to see every screen in section 6.

6 Guided tour of every screen

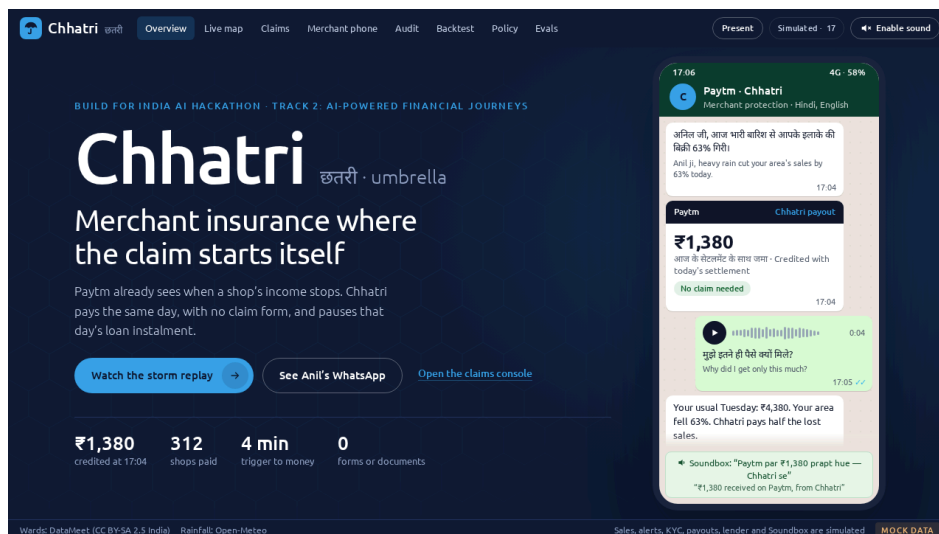
These screens were captured from the static mock with all 14 flags on. The console pictures are 1280 by 720 pixels (a projector). The phone pictures are 390 by 844 (a phone). Each picture says what it shows and what to do. Times are simulated clock times of the demo day, Tuesday 19 August 2025.

The console pages are: Overview, Live map, Claims, Merchant phone, Audit, Backtest, Policy, Evals (the menu across the top). The small clock bar under the menu moves the demo day.

The console (1280 by 720)

6.1 Overview

Menu: **Overview** · the front page

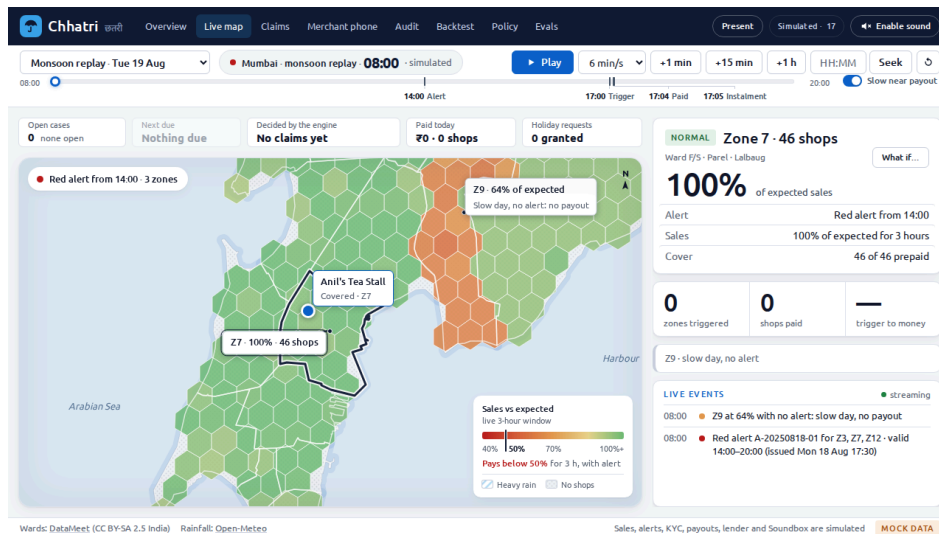


The front door. It says what Chhatrri is, shows the headline results (₹1,380 credited at 17:04, 312 shops paid, 4 minutes from trigger to money, 0 forms), and a phone with the WhatsApp chat Anil sees.

1. Press **Watch the storm replay** to jump to the storm on the live map.
2. Press **See Anil's WhatsApp** to see his chat.
3. Press **Open the claims console** to go to the officer queue.
4. In the header, the chip **Simulated** opens the provider panel (6.9). **Enable sound** lets the Soundbox voice play; click it once.

6.2 Live map, loaded at 08:00

Menu: Live map

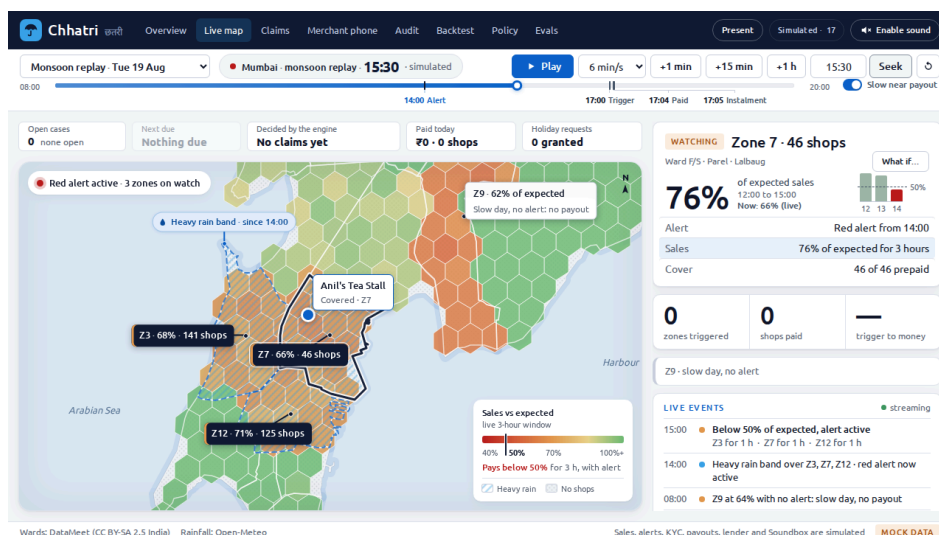


Mumbai ward by ward, before the storm. Each hexagon is a patch of shops. Green means sales are normal. The bar at the top is the demo clock.

1. Pick a day in the **scenario** list at the top left. Choose **Monsoon replay**. The clock reads *Mumbai · monsoon replay · 08:00 · simulated*.
2. Type a time such as 15:30 in the box and press **Seek** to jump there. Press **Play** to run it; the speed list sets how fast (6 min/s means 6 simulated minutes per second).
3. **+1 min**, **+15 min** and **+1 h** step the clock. The round arrow resets.
4. The marks under the bar are the story: 14:00 alert, 17:00 trigger, 17:04 paid, 17:05 instalment.
5. **Slow near payout** (on by default) slows the clock from 16:58 to 17:06 so you can watch.

6.3 The storm builds

Live map · seek to 15:30



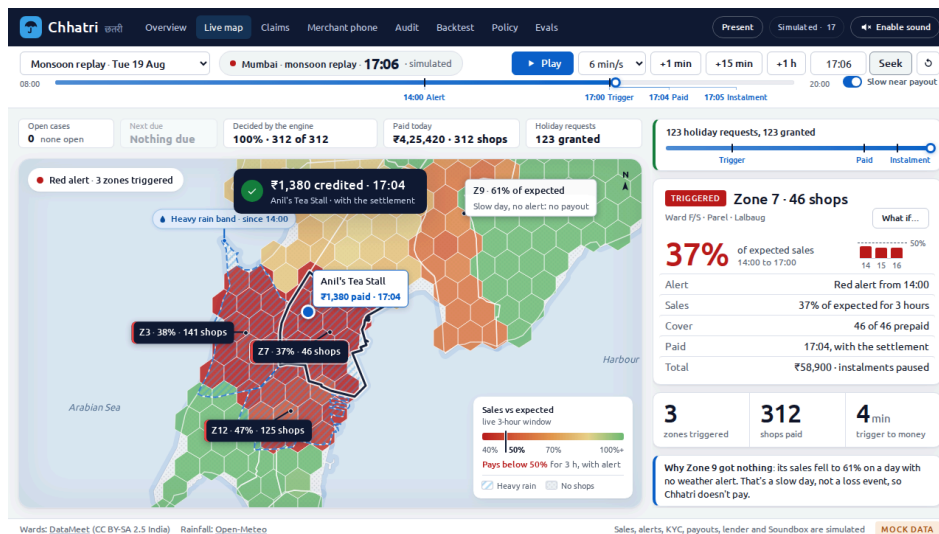
The red rain alert has been on since 14:00 for zones Z3, Z7 and Z12. Their sales are falling, so their hexagons turn amber and then red. Zone Z7 shows *Watching*.

1. Seek to **15:30**.
2. Look at the three zone labels on the map. A label such as **Z7 · 37% · 46 shops** reads: sales are 37% of expected, 46 shops.

3. Look at Z9 on the east side. It dips too, but there is no alert there.

6.4 The trigger fires, and the zone panel

Live map · seek to 17:06

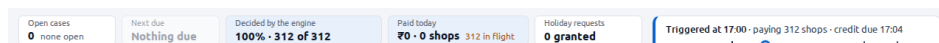


At 17:00 the drop has held below 50% for 3 hours during the alert, so the trigger fires for Z3, Z7 and Z12. The three tiles at the bottom right read 3 zones triggered, 312 shops paid, 4 minutes from trigger to money. A green note on the map says Anil's Tea Stall got ₹1,380 at 17:04.

1. Seek to **17:06** (or press Play from 14:30 and wait).
2. The card on the right is the **zone panel** for Zone 7: alert, sales (37% of expected for 3 hours), cover (46 of 46 prepaid), paid (17:04, with the settlement), total ₹58,900, instalments paused. Click another zone label to change the card.
3. Read the blue note at the bottom: *Why Zone 9 got nothing*. Its sales fell to 61% with no alert. A slow day is not a loss event, so Chhatrri doesn't pay.
4. The strip above the map is the **ops strip** (6.5). The bar above the zone panel is the **moment card** (6.6).

6.5 The ops strip

Top of the Live map and Claims · flag h8_ops_strip

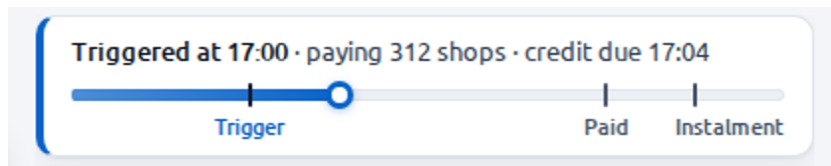


Five real counts, taken from the demo database: **Open cases**, **Next due** (the case whose 24-hour answer is nearest), **Decided by the engine**, **Paid today** and **Holiday requests**. This picture is from 17:01, when 312 payouts are still "in flight".

1. Seek to 17:01 and watch **Paid today** read ₹0 with "312 in flight".
2. Seek to 17:06 and see it fill in.
3. Click a tile for a small pop-up with the detail.

6.6 The moment card

Right of the Live map · flag `console_polish`

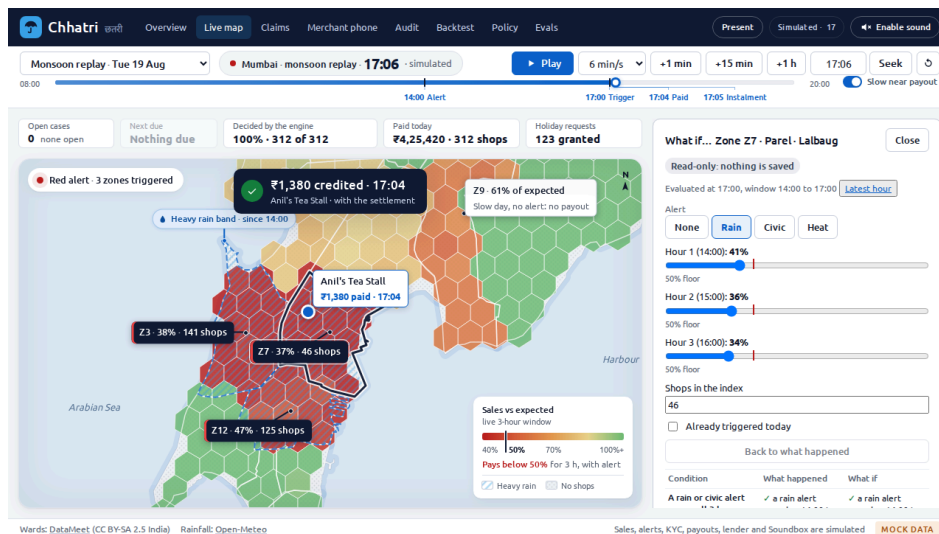


A small timeline from trigger to money. At 17:01 it reads *Triggered at 17:00 · paying 312 shops · credit due 17:04*. Once paid it turns green: *123 holiday requests, 123 granted*.

1. Seek to 17:01, then press **+1 min** four times and watch the dot move along Trigger, Paid, Instalment.

6.7 The what-if drawer

Zone card · button **What if...** · flag `h24_whatif`



A read-only calculator for judges. It re-runs the real trigger rule for Zone 7 with numbers you change. Nothing is saved and nobody is paid.

1. Click **What if...** on the zone card.
2. Choose the alert type: None, Rain, Civic or Heat.
3. Drag the three hour sliders (sales as a percent of expected). The red mark on each is the 50% floor.
4. Change **Shops in the index** (the rule needs at least 20).
5. Read the table at the bottom, which shows each condition for *What happened* and *What if*. Press **Back to what happened** or **Close** to leave.

6.8 Presenter mode

Header button **Present** · flag `console_polish`

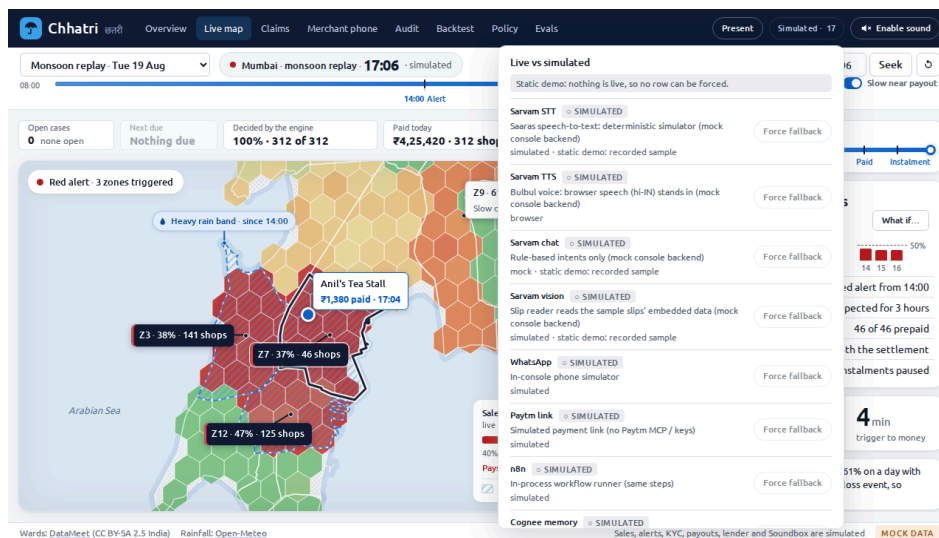


A bigger-type view for a projector. The header gets a **Keys** button, and the replay controls fold into a tidy row.

1. Press **Present** in the header. It stays pressed while the mode is on.
2. Press **Keys** to see the keyboard shortcuts for the presenter.
3. Press **Present** again to go back to the normal size.

6.9 Provider panel

Header chip **Simulated** (or Live) · flag `x6_provider_panel`



The list of every outside service and whether it is live, simulated or fallback. It is how the project proves that nothing simulated is shown as live. The top line says *Static demo: nothing is live, so no row can be forced.*

1. Click the chip in the header that reads **Simulated** · 17.
2. Read each row: name, label, a one-line reason (for example *Slip reader reads the sample slips' embedded data*).
3. With a real backend the **Force fallback** button switches a live service off, so you can show the fallback on purpose.
4. Press **Esc** to close.

6.10 Claims queue and the case panel

Menu: **Claims** (a red number counts open cases)

Wards: DataMeet (CC BY-SA 2.5 India) Rainfall: Open-Meteo

Sales, alerts, KYC, payouts, lender and Soundbox are simulated **MOCK DATA**

Where the claims officer works. On the left is the queue. On the right is the open case: here **C-2291**, a dispute. Anil said "my loss was bigger". The panel shows the engine's decision (₹1,380, formula, rule version), a line saying one more point of drop would have added about ₹22, the lender's answer on the instalment, and the evidence chart.

1. To get a case: open **Merchant phone**, tap the Hindi voice chip that means *my loss was bigger*, then follow the link **C-2291**.
2. Read the grey decision box. It says **The amount cannot change**.
3. Type an optional note, then press **Confirm payout** (keeps the payout) or **Reject dispute** (closes it). The merchant is told either way.
4. For a REFERRED slip case (a slip with the wrong name) the officer presses **Approve**; the approval re-runs every hard check and then pays ₹1,500.

6.11 Audit log with Verify chain

Menu: **Audit**

Wards: DataMeet (CC BY-SA 2.5 India) Rainfall: Open-Meteo

Sales, alerts, KYC, payouts, lender and Soundbox are simulated **MOCK DATA**

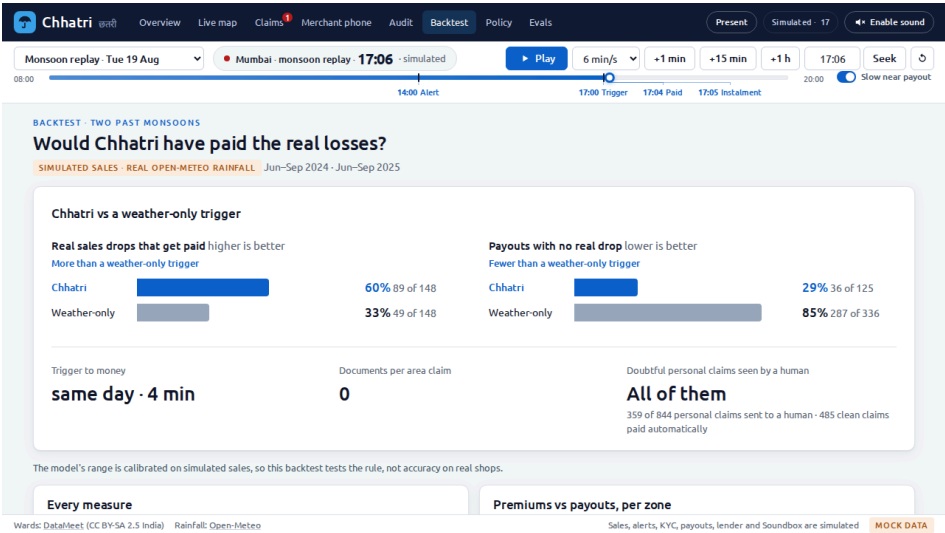
A tamper-evident log. Each entry stores a fingerprint (a sha256 hash) of the one before it, so changing any line breaks the chain.

1. Press **Verify chain**.

- 2. A green note reads *Chain valid · 22 entries · head 27a12c0c93...* (your count may differ).
- 3. Type in the filter box to find an action, an actor or a subject, for example `instalment`.
- 4. Read the rows from the bottom up: trigger, decision, payout, holiday request, lender answer, message.

6.12 Backtest

Menu: Backtest

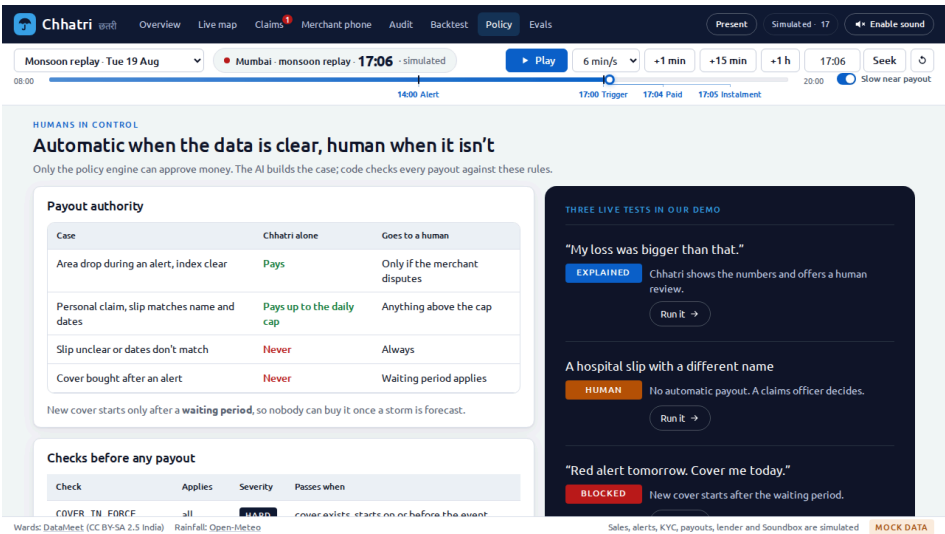


Would Chhatrri have paid the real losses of the 2024 and 2025 monsoons? It compares Chhatrri with a weather-only trigger. It is labelled *simulated sales · real Open-Meteo rainfall*.

- 1. Left chart: real sales drops that get paid. Chhatrri paid 89 of 148 (60%); a weather-only trigger paid 49 (33%).
- 2. Right chart: payouts with no real drop. Chhatrri 36 of 125 (29%); weather-only 287 of 336 (85%).
- 3. Below: trigger to money (same day, 4 minutes), documents per area claim (0), doubtful personal claims seen by a human (all of them).
- 4. Scroll down for every measure and premiums against payouts per zone. Read the line under the chart: the model tests the rule, not accuracy on real shops.

6.13 Policy

Menu: Policy



The payout-authority table (who may decide what) and the checks that run before any payout. On the right are the three live tests of the demo: EXPLAINED, HUMAN and BLOCKED.

1. Read the table: green means Chhatri pays alone, red means never alone.
2. Press **Run it** on a test card to load that scenario.
3. Scroll to **Checks before any payout**: each check, where it applies, whether it is HARD, and when it passes.

6.14 Evals

Menu: Evals · flag `h25_evals`

The screenshot shows the Chhatri Evals interface. At the top, there's a navigation bar with tabs: Overview, Live map, Claims, Merchant phone, Audit, Backtest, Policy, and Evals. Below the navigation bar is a timeline showing a monsoon replay for Tuesday, 19 Aug, with a simulated event at 17:06. The main content area is titled 'AI evaluation' and shows a note: 'Synthetic data only. Results on generated slips and written questions say little about real merchants or real hospital paper. The plan: docs/04-engineering/ai-evaluation-plan.md'. Below this, there are six evaluation suites, each with a status of 'NOT MEASURED' and 'no run stored':

- S1 Intent routing
- S2 Guard red-team
- S3 Ask end to end
- S4 Slip reading and the gate
- S5 Voice
- S6 Chains and labels

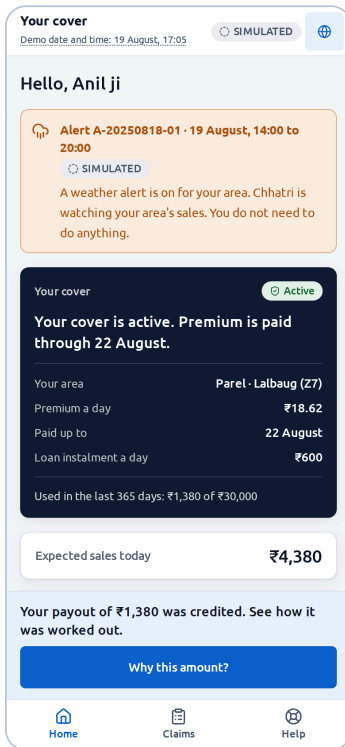
At the bottom, there are footnotes: 'Wards: DataMeet (CC BY-SA 2.5 India) Rainfall: Open-Meteo' and 'Sales, alerts, KYC, payouts, lender and Soundbox are simulated'. A 'MOCK DATA' label is also present.

The AI evaluation page. It shows six suites (intent routing, guard red-team, Ask end to end, slip reading and the gate, voice, chains and labels). No run is stored, so every suite honestly reads **NOT MEASURED** and no number is shown.

1. Open the page and read the yellow note: synthetic data only.
2. After someone runs `make evals` and stores a run, each card shows its measured result instead.

The merchant mini-app (390 by 844)

The mini-app is "Chhatri in Paytm for Business", a phone-sized app inside the console (page **Merchant phone**) and also full screen at `/merchant/S-0142/app`. The first merchant is Anil (S-0142), shown at 17:05 on the storm day. Three tabs at the bottom (Home, Claims, Help) move between the main areas; the bar above them always suggests the next step. The globe icon at the top right changes language. On the static demo a blue banner says *This is the static demo*; press **Got it** to fold it away (it is folded away in the pictures below).

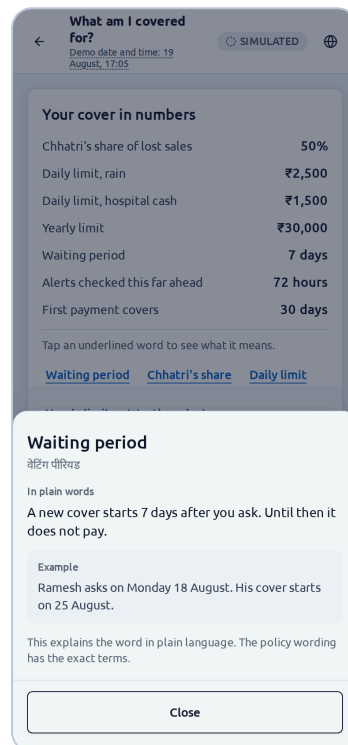
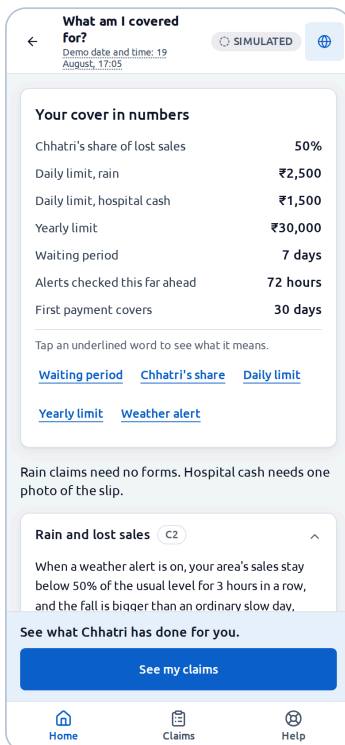


6.15 Home

Tab Home

Anil's front screen. It greets him, shows the weather alert for his area, and the status of his cover: active, premium paid through 22 August, ₹18.62 a day, ₹600 loan instalment, used ₹1,380 of ₹30,000 this year.

1. Read the amber alert box. It says watching, you do not need to do anything.
2. Read the dark cover card: area, premium a day, paid up to, loan instalment.
3. Press the blue bar at the bottom, **Why this amount?**, to see how ₹1,380 was worked out.

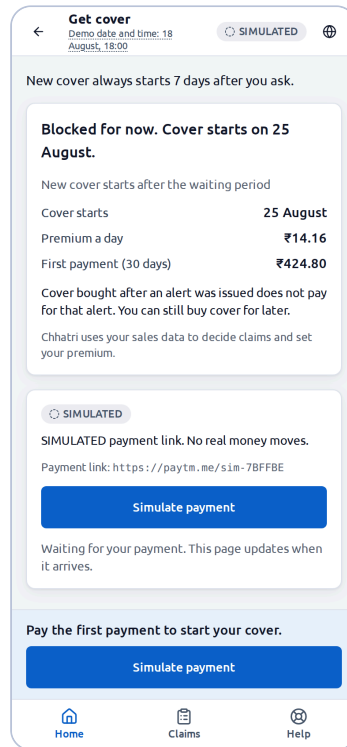
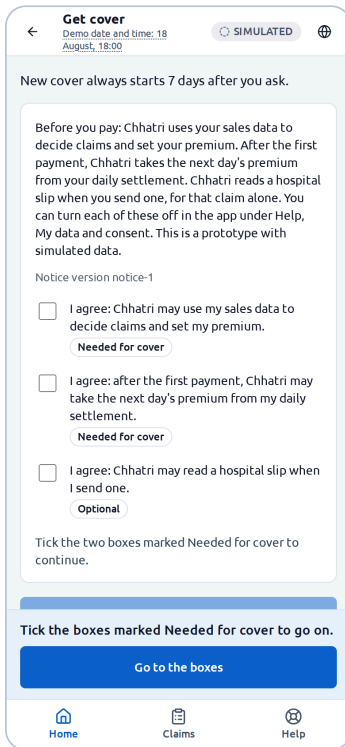


6.16 Coverage, and the jargon lens

Home · What am I covered for?

The cover in numbers: Chhatrati pays 50% of lost sales, up to ₹2,500 a day for rain and ₹1,500 a day for hospital cash, ₹30,000 a year, a 7-day waiting period, 72 hours of alert look-ahead and a first payment that covers 30 days. Underlined words are explained on tap: the **jargon lens** (a sheet that explains a word in plain language, with one example and the other language).

1. Scroll through **Your cover in numbers** and the examples.
2. Tap an underlined word such as **Waiting period**. A sheet opens with *In plain words*, an example and the Hindi word.
3. Press **Close** or the Esc key to go back.

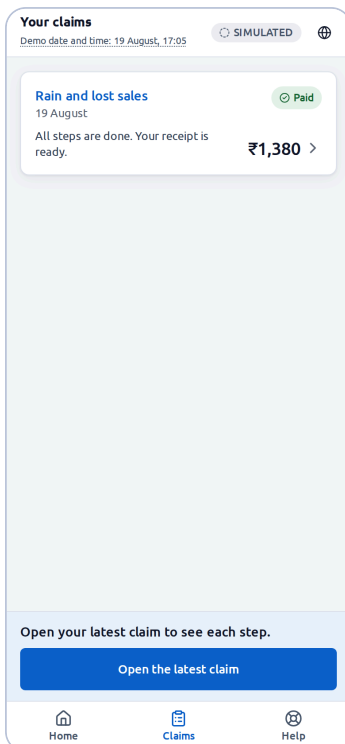


6.17 Buy cover, with consent

Tab **Home** · **Get cover** · shown for Ramesh (S-0907) on the *buy cover* scenario, Monday 18 Aug 18:00

Cover can only start after a 7-day waiting period, so nobody can buy it once a storm is forecast. The left picture is the form. Before paying the merchant ticks the boxes marked **Needed for cover** (use sales data to decide claims; take the first payment and then the daily premium from the settlement). A third box, reading a hospital slip, is optional. The right picture is the answer after pressing **Check price and start date**.

1. Read the notice, then tick the two **Needed for cover** boxes. The button stays grey until you do.
2. Press **Check price and start date**.
3. The answer says *Blocked for now. Cover starts on 25 August*. with the price: ₹14.16 a day, first payment ₹424.80 for 30 days.
4. Press **Simulate payment** on the simulated Paytm link. No real money moves.

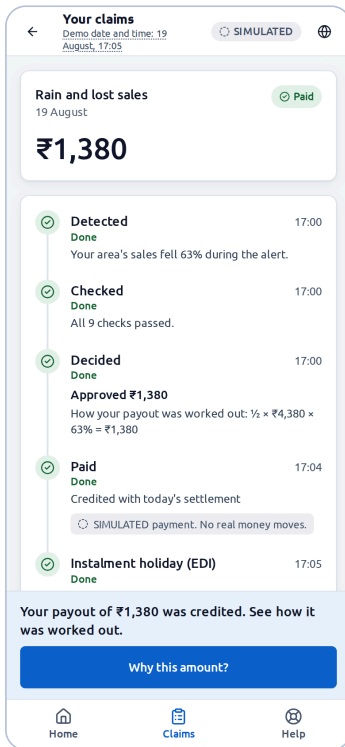


6.18 Claims tracker

Tab **Claims**

The list of Anil's claims. One card: *Rain and lost sales*, 19 August, Paid, ₹1,380, "All steps are done. Your receipt is ready."

1. Tap the claim card to open it.
2. If the list is empty it says so and tells you the next step.

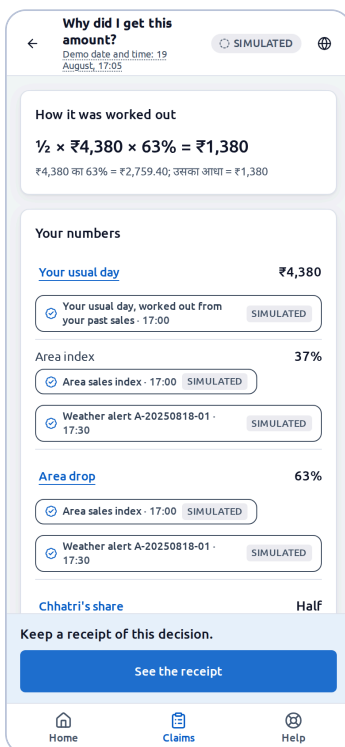


6.19 Claim detail

Claims · the claim card

A five-step tracker: **Detected** (17:00), **Checked** (17:00, all 9 checks passed), **Decided** (17:00, approved ₹1,380), **Paid** (17:04, credited with today's settlement) and **Instalment holiday** (17:05, the lender's answer). Paid and holiday steps carry a SIMULATED label.

1. Read each step top to bottom: state, simulated time and one line of reason.
2. Scroll to the end to see the lender's answer: the ₹600 instalment moves to the end of the loan with no penalty.
3. Press **Why this amount?** at the bottom.

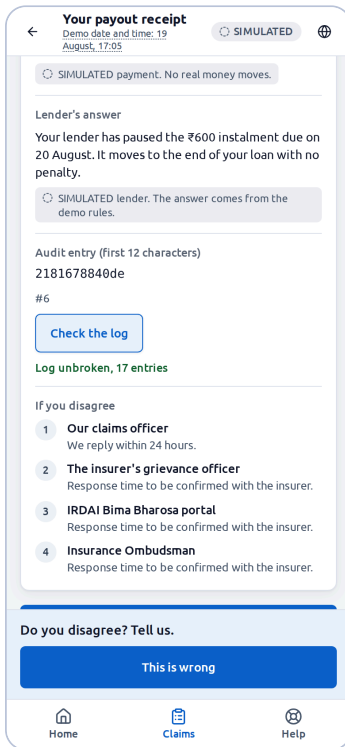


6.20 Why this amount

Claim detail · **Why this amount?**

The sum: $\frac{1}{2} \times ₹4,380 \times 63\% = ₹1,380$ in English and in Hindi. Each number sits on a row with a **source badge** (a small label that says where the number came from and when, for example Area sales index · 17:00 · SIMULATED).

1. Read the formula at the top.
2. Read **Your numbers**: your usual day ₹4,380, area index 37%, area drop 63%, Chhattri pays half.
3. Tap a badge to see its source.
4. At the bottom a line says what would have changed the outcome, written by the engine, not by a model.

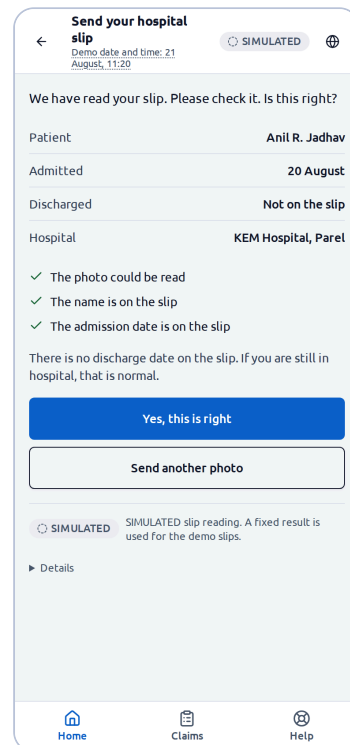
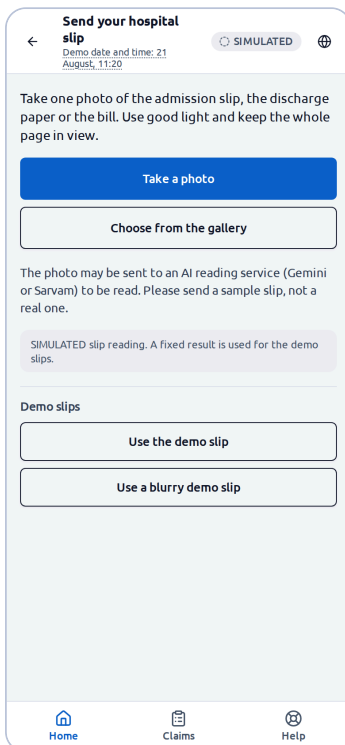


6.21 Receipt

Why this amount · [See the receipt](#)

A payout receipt: decision id, rules version, formula, sources and the first 12 characters of the audit entry. It can be printed or saved as a PDF.

1. Press **Check the log**. It answers *Log unbroken, N entries*.
2. Read **If you disagree**: your claims officer (24 hours), then the insurer's grievance officer, then IRDAI's Bima Bharosa portal, then the Insurance Ombudsman.
3. Press **This is wrong** to file a complaint (6.25).

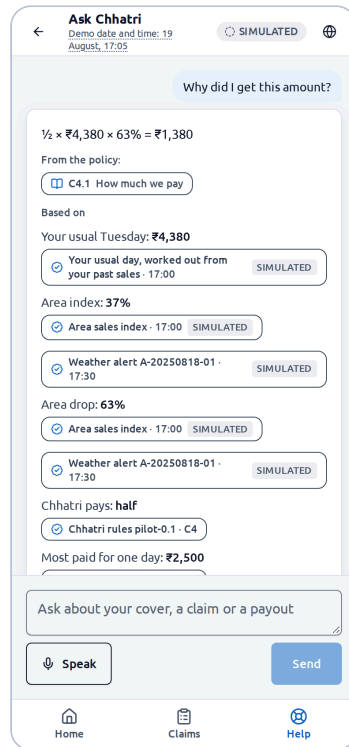
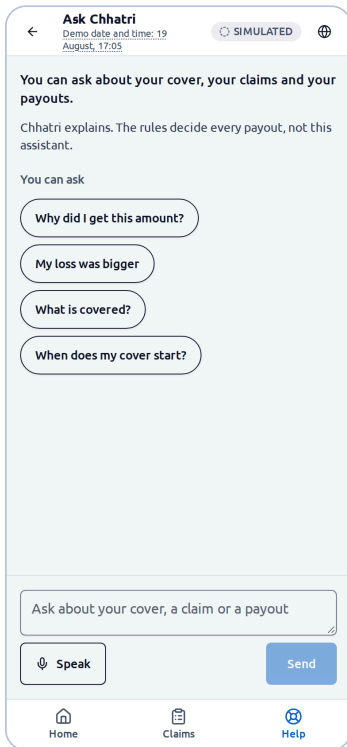


6.22 Slip pre-check

Tab **Help** · **Send your hospital slip** · scenario *illness*, Thursday 21 Aug 11:20

The personal claim. Chhatri noticed Anil's shop had no sales yesterday and asked if he is well. Now he sends one photo of the hospital slip. The app reads it and shows what it found before any check runs. Anil must confirm.

1. Press **Take a photo** or **Choose from the gallery**. For the demo, press **Use the demo slip** (or **Use a blurry demo slip** to see a bad photo handed back).
2. If asked, tick the consent for reading the slip.
3. Read the fields: patient Anil R. Jadhav, admitted 20 August, hospital KEM Hospital, Parel, and the three-line checklist (photo readable, name on the slip, admission date on the slip).
4. Press **Yes, this is right** to open the claim (₹1,500 approved), or **Send another photo**.

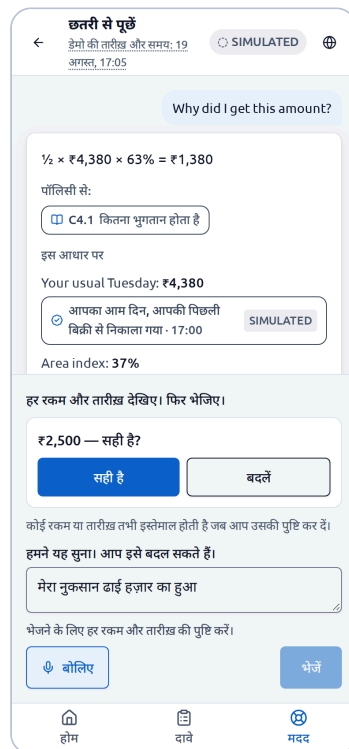
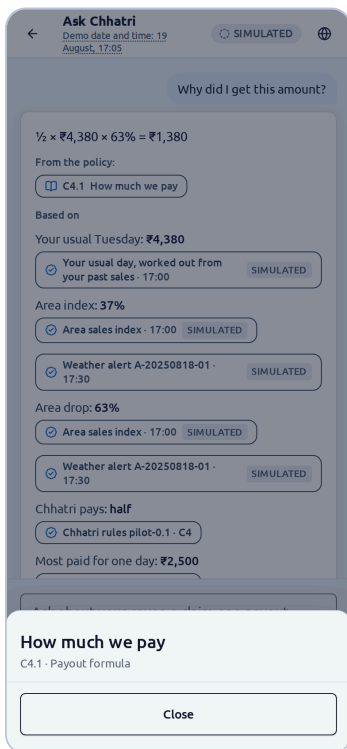


6.23 Ask Chhatri, with citations

Tab Help · Ask Chhatri

A question box. The answer is built from the engine's own facts and cites the policy clause it relies on. At the bottom of an answer a small label says what wrote it (SIMULATED, FALLBACK or LIVE).

1. Tap an example question, such as **Why did I get this amount?**, or type your own and press **Send**.
2. Read the answer: formula, then *From the policy* with a clause chip such as **C4.1 How much we pay**, then *Based on* with the numbers and their source badges.
3. Tap the clause chip to read the clause in a sheet (next picture).
4. If a message looks like a scam (asks for an OTP or an advance fee), a warning banner appears.

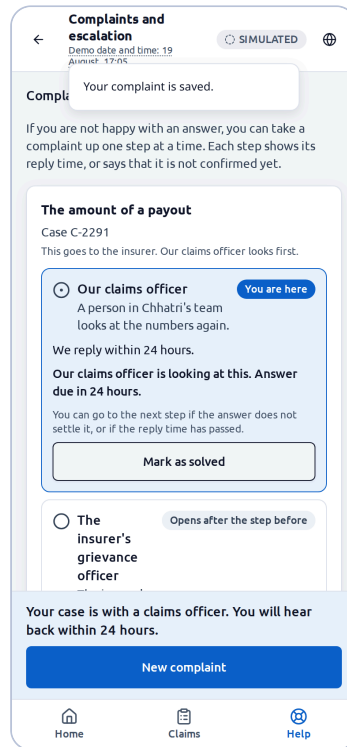
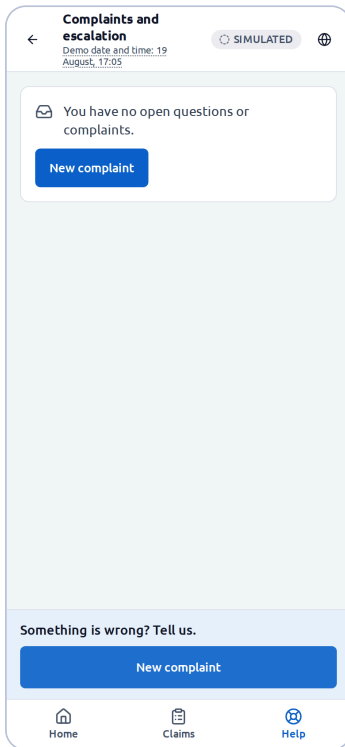


6.24 Voice chips

Ask Chhatri · **Speak** (shown in Hindi) · flag n4_voice

The left picture is the clause sheet. The right picture shows voice: the app listens, writes what it heard, and when it hears an amount it asks Anil to confirm it with a chip, **₹2,500, is that right?** with buttons **Yes** and **Change**. **Send** stays grey until he confirms.

1. Press **Speak**. A first-use notice says the voice goes to a speech service or the browser; press **Continue**.
2. Speak, then press the button again to stop.
3. Check the text. Tap **Yes** on the amount chip, or **Change** to correct it.
4. Press **Send**. If you edit the text to add a new amount, you must confirm again.

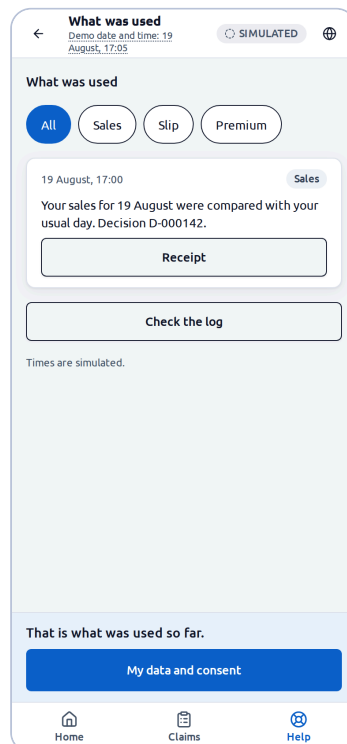
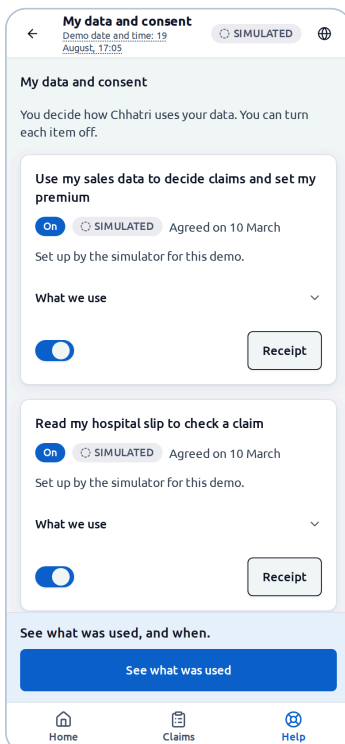


6.25 Grievances

Tab **Help** · **Complaints and escalation** · flag `n5_grievances`

Where a merchant complains. With nothing filed, it says so. After filing it shows one ladder: the claims officer (you are here, we reply within 24 hours), then the insurer's grievance officer, then IRDAI's Bima Bharosa portal, then the Insurance Ombudsman. The outside steps are labelled SIMULATED or "file it yourself".

1. Press **New complaint**.
2. Pick a topic, for example **The amount of a payout**, then write a few words (up to 500 characters).
3. Press **Send**. The ladder appears with the case number.
4. Press **Mark as solved** to close it.

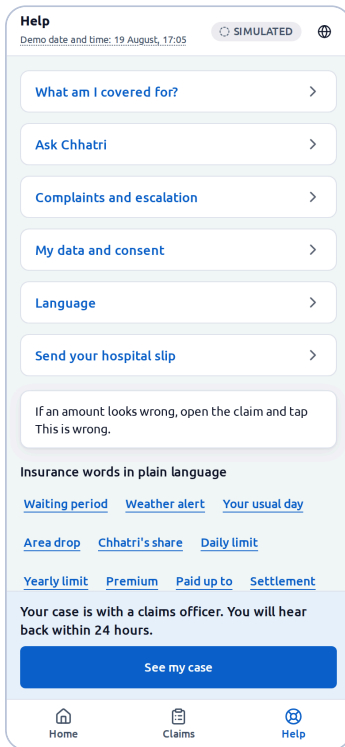


6.26 Consent centre and activity

Tab **Help** · **My data and consent** · flag `n6_consent`s

Consent is the merchant's permission to use data for one purpose. Each purpose has a switch (use my sales to decide claims, read my hospital slip), the date it was agreed, and a **What we use** note. The right picture is the activity log: what was used, for what and when.

1. Turn a switch off. A sheet explains the effect, for example that Chhatr stops reading slips. Confirm or keep it.
2. Press **See what was used and when** to open the activity log. Filter by Sales, Slip or Premium.
3. Press **Check the log** to verify nothing was changed.
4. If a slip is held, press **Erase this slip** to delete it on request.

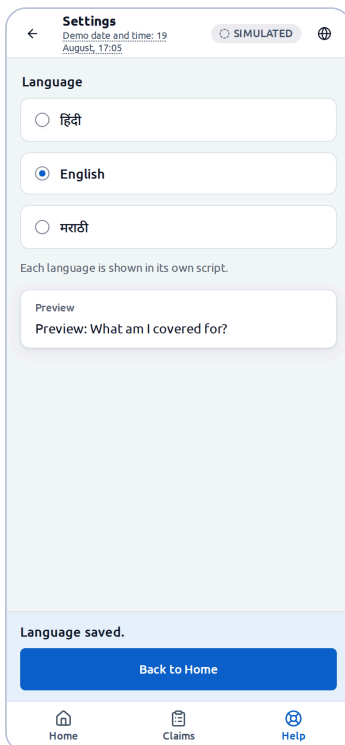


6.27 Help

Tab Help

One list of everything else: what am I covered for, Ask Chhattri, complaints, my data and consent, language, and send your hospital slip. A line explains that if an amount looks wrong you open the claim and tap **This is wrong**. Insurance words are listed with the jargon lens.

1. Tap a row to open it. Back returns here.
2. Tap a word such as **Weather alert** to see it explained.



6.28 Language switch, in Hindi

Help · **Language**, or the globe icon

The app speaks Hindi, English and (with n8_marathi) Marathi. Each language is written in its own script. The right picture is Home after choosing Hindi: the same cover card, the same numbers, in Devanagari.

1. Tap **हिन्दी**. A preview line shows how it will read.
2. Press **Back to Home**. The tabs now read होम, दावे, मदद.
3. Switch back with the same screen.

7 The three-minute demo story

This is the short cut of the demo from the runbook. One person narrates and a second person clicks. Seek the console to 14:30 on the Live map before you start, with the monsoon scenario loaded and the replay paused. For the longer seven-minute story (illness, the mismatched slip, the blocked cover purchase, audit, policy and backtest) see [docs/DEMO.md](#).

Time	What you say	What you do	What the screen shows
0:00-0:30 · the hook	"Anil runs a tea stall in Parel. On a day of heavy rain his sales fall by more than half, and his ₹600 loan instalment is still cut. Getting paid for such a day has typically taken 30 to 60 days. With Chhatri the claim starts itself and he is paid the same evening."	Nothing. Console parked on the Live map.	The opening slide or the Overview.
0:30-1:15 · the trigger	"This is Mumbai, ward by ward. A red rain alert from 14:00 for three wards. Watch their sales." Then: "The drop held for three hours across all 46 shops in Zone 7. No one filed anything." And: "A slow day is not a loss event."	Press Play at 6 min/s. About 33 seconds later pause near 17:06. Click Z7 .	Zones go amber then red. 3 zones triggered, 312 shops paid, 4 min. Z7 card: 37%, 46 of 46 prepaid, ₹58,900. Z9 reads 61%, slow day.
1:15-1:31 · Anil's phone	"₹1,380, credited with today's settlement, and the Soundbox says it out loud. In his app he follows the claim: detected, checked, decided, paid, and a holiday on tomorrow's ₹600 instalment that Chhatri asks his lender for. The lender decides. Here it is simulated."	Open Merchant phone , then tap the latest claim in the mini-app.	The ₹1,380 card, the Soundbox line, then the five-step tracker with SIMULATED labels.
1:31-1:45 · why	"His usual Tuesday is ₹4,380, his area fell 63%, and Chhatri pays half the lost sales. Every number says where it came from."	Tap Why this amount? , then one source badge.	$\frac{1}{2} \times ₹4,380 \times 63\% = ₹1,380$ with badges.
1:45-2:05 · Ask	"Anil can also ask in his own words. The answer cites the policy clause and says what wrote it."	Help, then Ask Chhatri . Tap an example question.	An answer with a clause chip and a label such as SIMULATED.
2:05-2:40 · who decides	"Half of ₹4,380 times 63% is ₹1,380. The AI builds the case; code decides the money."	Open the Policy page.	The payout-authority table.
2:40-3:00 · the close	"Sales, alerts, KYC, payouts, the lender, WhatsApp and the Paytm link are simulated and labelled on screen."	Nothing.	The closing slide.

IF THE CLOCK RUNS LATE

Drop items in this order: the Z7 click and the Zone 9 line, the tracker, the badge tap, the Ask step, the Policy click. **Never drop the closing disclosure sentence.**

Golden numbers to check against the screen

Fact	Value
Replay day	Tuesday 19 August 2025, 08:00 to 20:00 (simulated)
Alert	A-20250818-01, red rain, zones Z3, Z7, Z12, issued Monday 18 Aug 17:30, valid 14:00 to 20:00

Fact	Value
Indices at 17:00	Z7 37% · Z3 38% · Z12 47% · Z9 61% (slow day, no alert)
Timeline	Decisions 17:00 · credits, WhatsApp and Soundbox 17:04 · lender asked, instalments paused 17:05
Totals	3 zones triggered · 312 shops paid · 4 min trigger to money · Z7 total ₹58,900
Anil	Usual Tuesday ₹4,380 · payout ₹1,380 · instalment ₹600 paused by his lender
First case	C-2291 after a fresh load of a scenario

If the console ever shows something different, reload the scenario from the list, and if it still differs, run `make demo-check` before going on.

8 Features list

Each feature has an id. **N** is a new feature, **X** is a fix to something that already existed, **H** is an idea adopted from another project. "Flag" is the switch from section 5. "none" means it is always on.

N: new features

Id	In plain words	Flag
N1	The merchant mini-app: a phone-sized app with home, coverage, buying cover, claims, receipt and help.	n1_miniapp
N2	Ask Chhatrī: ask a question in your own words and get an answer that cites the policy.	n2_ask_chhatrī
N3	Slip reading with a pre-check: the app shows what it read and you confirm before anything is checked.	n3_slip_precheck
N4	Real Hindi voice: speak a question and hear an answer.	n4_voice
N5	Complaint ladder with response clocks: who to complain to next, and by when.	n5_grievances
N6	Consent centre: see, turn off and log each permission; erase your slip.	n6_consent
N7	A static build of the console that runs with no backend. The build exists; it has not been put on a public address.	none
N8	Marathi as a third language (a draft, text only).	n8_marathi

X: fixes

Id	In plain words	Flag
X1	The frontend tests pass, every time, even on a busy laptop.	none
X2	A bad claim is never stored: the "usual day" amount must be a properly rounded published value.	none
X3	A missing zone price now fails loudly instead of quietly charging the ₹2 minimum.	none
X4	The lender decides the instalment holiday. Chhatrī only asks. Before this fix the pause was unconditional.	x4_lender_request
X5	Every route answers an unknown merchant id with a clean "not found".	none
X6	Provider panel with a fallback switch: see which services are live and force a fallback.	x6_provider_panel
X7	An honest-wording test: no screen claims a simulated thing is live.	none
X8	No loan offers to a merchant in distress, and a cap on messages.	x8_distress_guard

H: ideas adopted from other projects

Id	In plain words	Flag
H1	Claim tracker: Detected, Checked, Decided, Paid, Instalment holiday.	n1_miniapp
H2	"Why this amount" with source badges.	none (view in n1_miniapp)
H3	A payout receipt you can print.	none
H4	Honest-wording test (same as X7).	none
H5	The merchant confirms what was read from the slip.	n3_slip_precheck
H6	A public demo with no login (the static build; not published yet).	none
H7	A provider panel that shows how the AI services are doing, with a fallback switch.	x6_provider_panel
H8	Real operations counts taken from the database, never projections.	h8_ops_strip
H9	No loan offers in distress, plus a message cap.	x8_distress_guard
H10	A deterministic core: the AI has no authority over money.	none
H11	Bilingual message templates instead of model translation.	none (Marathi: n8_marathi)
H12	Visible, measured test counts in the README.	none
H13	Source badges: every rule, number and clause says where it came from, and when.	none
H14	A counterfactual in every explanation: what would have changed the outcome, written by the engine.	none
H15	Slip pre-check: document type, checklist, and "ask, do not assume" when the reading is unsure.	n3_slip_precheck
H16	Defence against orders hidden in slips and chat.	none
H17	Ask Chhatri cites policy clauses; every number comes from the engine.	n2_ask_chhatri
H18	Voice confirmation chips for amounts and dates.	n4_voice
H19	Scam-message warning in chat (OTP requests, advance fees, "guaranteed claim", urgency).	n2_ask_chhatri
H20	Jargon lens: tap an insurance word for a plain explanation.	n1_miniapp
H21	Next-step bar: no screen or reply is a dead end.	n1_miniapp
H22	Grievance ladder with response clocks and a "who owns this complaint" router.	n5_grievances
H23	Consent activity log and "forget my slip".	n6_consent
H24	What-if drawer for judges (read-only).	h24_whatif
H25	Published AI evaluation: a number is shown only once it is measured.	h25_evals
H26	Every AI reply carries its mode (live, simulated, fallback), provider and reason.	none

Console-only flags: `console_polish` (Presenter mode, the moment card and projector type) and `h8_ops_strip`, `h24_whatif`, `h25_evals`, `x6_provider_panel`. All 14 flags are listed with their effect in section 5.

9 Troubleshooting, FAQ and glossary

Troubleshooting

Symptom	What to do
Numbers differ from this manual	Reload the scenario from the list at the top left (it is deterministic). If they still differ, run <code>make demo-check</code> .
A page or button is missing	Its flag is off. Set the flag in both <code>CHHATRI_FEATURES</code> and <code>VITE_FEATURES</code> (section 5) and restart. A flagged route answers 404 when off.
Page is blank or reset after pressing reload in the static mock	The mock backend lives in the page, so a reload restarts it. Load the scenario again and use the menu links, not reload.
<i>Reconnecting pill</i>	Wait. The stream resumes by itself. If it lasts over 10 seconds, reload the page; the state is kept by the backend.
No sound	Press Enable sound in the header once. Every voice bubble also has a play button.
Phone shows nothing after a slip or an officer approval	The clock is paused and money arrives 4 simulated minutes later. Press Play , or press +15 min .
Replay stuck	Pause, press +15 min , or seek to the time you need.
Map has no street tiles	Nothing to fix. Without internet or a tile key the console draws the ward outlines on a plain background.
<code>make demo-check</code> fails many steps	Check that <code>n3_slip_precheck</code> and <code>n6_consent</code> are not on, and that <code>x4_lender_request</code> is.
After saving a backend file, the replay and cases vanish	<code>make dev</code> restarts the backend when a file changes, which clears everything held in memory. Reload the scenario and seek again.
Console does not load with <code>make up</code>	Run <code>docker compose ps</code> and <code>docker compose logs backend</code> . As a fallback run <code>make dev</code> (console on port 5173).
Clock holds at 17:00 for 30-60 seconds (n8n live)	Expected: 312 payout runs go through n8n one after another. To avoid it, set <code>CHHATRI_STACK_N8N_URL=</code> (empty) in <code>.env</code> .
<code>/api/preflight</code> shows an item that is not ok	An artefact is missing. Run <code>make data</code> (slow) or restore the committed files.

Questions a newcomer asks

Is Chhatri a real insurance product?

No. It is a hackathon prototype. No insurer, lender or Paytm team has agreed to take part, and nothing here is an approved product.

Does real money move?

No. Payouts, the lender and the payment link are simulated and labelled.

Who decides the money, the AI or the code?

The code. A policy engine reads `rules.yaml` and decides. The AI only builds the case and explains.

What happens when the AI is unsure or unavailable?

A doubtful slip goes to a claims officer, never to an automatic payout. If an AI service is not available, a simple template or simulator answers and the label says so.

How is the payout worked out?

For a storm, half of the lost sales: $\frac{1}{2} \times \text{the usual day} \times \text{the drop}$. For Anil, $\frac{1}{2} \times ₹4,380 \times 63\% = ₹1,380$. For illness, half of the usual day, capped at ₹1,500 a day.

Why did Zone 9 get nothing?

Its sales fell to 61% on a day with no weather alert. That is a slow day, not a loss event.

Does Chhatri pause a loan?

No. It asks the lender, under a rule agreed in advance, and the lender decides. In the demo the simulated lender says yes.

Why can a merchant not buy cover when a storm is forecast?

Cover starts after a 7-day waiting period, with a 72-hour look-ahead at alerts. Otherwise people would buy cover only on the eve of a storm.

Do I need API keys or an internet connection?

No. Without keys everything is simulated, and the static mock runs in the browser with no backend.

Which languages does it speak?

Hindi and English. Marathi is a draft, in text only, behind `n8_marathi`. Voice is Hindi and English.

Is any real person's data used?

No. Every merchant, slip and sales figure is synthetic. AI services are called only when that is declared.

How can I check that nobody changed the log?

Open the Audit page and press **Verify chain**, or press **Check the log** on a receipt in the mini-app.

Glossary

Term	Meaning
Alert	A weather warning from the India Meteorological Department with colours: Green, Yellow, Orange, Red. Chhatri looks at Red and Orange.
Audit log	A record of every step. Each entry holds the fingerprint of the one before, so a change is detected.
Backtest	A replay of past monsoons to see what the rules would have paid. Here it uses simulated sales and real rainfall.
Basis risk	The gap between what the trigger says and what the merchant really lost.
Case	An open dispute or doubtful claim, with a 24-hour answer clock, such as C-2291.
Claim	A request to be paid after a loss.
Consent	A permission to use your data for one named purpose. You can withdraw it.
Counterfactual	A line that says what would have changed the outcome.
EDI	Equated daily instalment: the daily loan repayment taken from a merchant's settlement.
EDI holiday	A pause of one instalment, which moves to the end of the loan. Chhatri asks; the lender decides.

Term	Meaning
Expected sales	What a shop normally sells at that hour or on that weekday, estimated by a model.
Feature flag	A switch that is off until you turn it on.
Hospital cash	A fixed daily amount for each day in hospital, whatever the bill.
IRDAI	The Insurance Regulatory and Development Authority of India. Its Bima Bharosa portal takes insurance complaints.
Jargon lens	Tap an insurance word to see a plain explanation.
KYC	Know Your Customer: the identity record of the merchant. A slip's name must match it.
Lender	The business that lent the merchant money. It decides about an EDI holiday.
Parametric	Insurance that pays when a set trigger happens, without measuring each shop's loss.
Payout	Money paid after a claim is approved.
Policy engine	The code that decides APPROVED, REFERRED or DECLINED. Only it can approve money.
REFERRED	A decision that sends the claim to a person.
Replay	A run of a past day, with a clock you control.
rules.yaml	The file that holds the policy rules and numbers.
Sales index	Actual sales divided by expected sales for a zone.
Settlement	The daily payment from Paytm to a merchant, after fees and loan deductions.
Slip	The hospital paper (admission or discharge) used as proof for a hospital claim.
Soundbox	The speaker on a merchant's till that announces payments.
Source badge	A small label that says where a number came from and when.
Trigger	The event that starts a claim.
Waiting period	The days after buying cover before it can pay. Here, 7 days.
Zone	A group of neighbouring shops in one area. The demo uses Z3, Z7, Z9 and Z12.

WHERE TO READ MORE

The documents index is `docs/README.md`. The demo script is `docs/DEMO.md`. How the parts fit is in `docs/ARCHITECTURE.md`, and every outside service is described in `docs/INTEGRATIONS.md`. Data credits: ward boundaries from DataMeet Mumbai BMC wards (CC BY-SA 2.5 India); weather data by Open-Meteo.com (CC BY 4.0).